PHYSICAL AND CHEMICAL SCIENCE

Energy and Work

Forms of energy, energy changes and transfers, tracking energy through systems, and saving energy



In this pack

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This pack belongs to: _____ **Class:** _____

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LEARN IT · C8 · ENERGY

Energy makes things happen

★☆☆ Grade 4

Name: Class: Date:

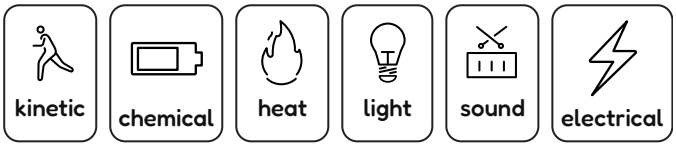
Learning goals

- Explain that energy is needed to make things work or move.
- Name forms of energy and describe how energy is converted and transferred.
- Track energy through a system and explain how people can save energy.

You need energy to run, to think and to grow. A bus needs energy to move, a fan needs energy to spin, and a lamp needs energy to shine. **Energy** is what makes things **work**, **move**, **change** or **happen**. Energy cannot be seen, but we can see what it does.

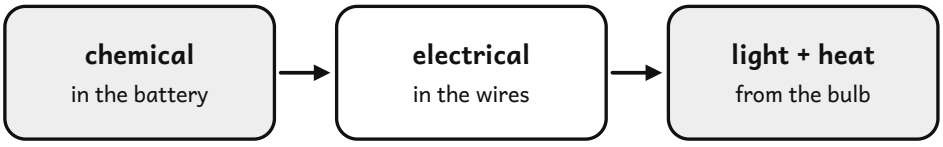
Forms of energy

Form	Examples
kinetic (movement)	a moving rickshaw, wind, a flowing river
stored (potential)	a book on a high shelf, a stretched rubber band, water behind a dam
chemical (stored)	food, wood, gas, petrol, batteries
heat (thermal)	a hot cup of cha, the Sun, a cooking stove
light	the Sun, a lamp, a firefly
sound	a dhol, a car horn, your voice
electrical	electricity flowing in wires



Energy changes form

Energy can be **converted** (changed) from one form to another. We show this with an **energy chain**:



A torch.

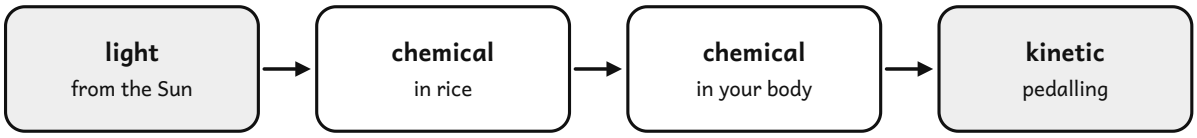
- **Electric fan**: electrical → kinetic (and a little sound and heat).
- **Solar panel**: light → electrical.
- **Your body**: chemical (food) → kinetic (movement) + heat.
- **Gas stove**: chemical → heat + light.

Energy moves from object to object

Energy can be **transferred** (passed on): a hot spoon passes heat to your hand; your foot passes kinetic energy to a football; a drum passes sound energy through the air to your ears.

Tracking energy through a system

Almost all our energy starts in the **Sun**. Follow it through a system: the Sun's light helps rice grow (stored as chemical energy). You eat the rice. Your muscles use the energy to pedal a bicycle (kinetic energy). Some energy is always lost as **heat**, so no machine turns all its energy into useful work.



Tracking energy from the Sun to a bicycle ride.

Energy that can run out

Coal, oil and natural gas are energy stores that took millions of years to form. Burning them gives us electricity and fuel, but they will **run out**, and they pollute the air. **Renewable** sources such as **sunlight**, **wind**, **flowing water** and **biogas** will not run out.

You can save energy

- Switch off lights, fans and TVs when no one is using them.
- Use LED bulbs: they give the same light with much less electricity.
- Walk, cycle or share transport for short trips.
- Cook with lids on pots so less heat escapes.

Bangladesh connection

Bangladesh has one of the largest **solar home system** programmes in the world. Millions of families in villages use small solar panels on their roofs to light their homes and charge phones.

Key words

energy	what makes things work, move or change
kinetic energy	energy of movement
stored energy	energy waiting to be used, such as in food or a stretched band
convert	change from one form to another
transfer	pass from one object to another
energy chain	a diagram showing how energy changes
renewable	will not run out
conserve energy	use less and waste less energy

FACT CHECK · C8 · ENERGY

Forms of energy

★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Name forms of energy.
- Complete energy chains.

1 Which form of energy? [6]

Example	Form of energy
a flowing river	
a plate of rice	
a ringing bell	
a stretched catapult	
sunshine	
a phone battery	

2 Complete the energy chains. [8]

Device	Energy in	Useful energy out
electric fan		
solar panel		
radio		
candle		

3 Sort the energy sources. [7]

SORT THESE

oil river water wind sunlight natural gas biogas coal

Renewable	Will run out

THINK IT THROUGH · C8 · ENERGY

Following the energy

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Track energy through systems.
- Explain energy losses.

1 **Draw an energy chain for a rickshaw puller, starting with the Sun.** [3]

Rickshaw energy chain

2 **An old filament bulb gets very hot. An LED bulb stays cool. Which one wastes more energy? Explain.** [2]

3 **Rubbing your hands together makes them warm. What energy change is happening?** [1]

4 **Why do we say almost all our energy comes from the Sun? Give two examples.** [2]

5 **Suggest two ways your school could save energy.** [2]

INVESTIGATE · C8 · ENERGY

Rubber band launcher

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Plan a fair test.
- Plot a line graph.
- Explain using stored and kinetic energy.

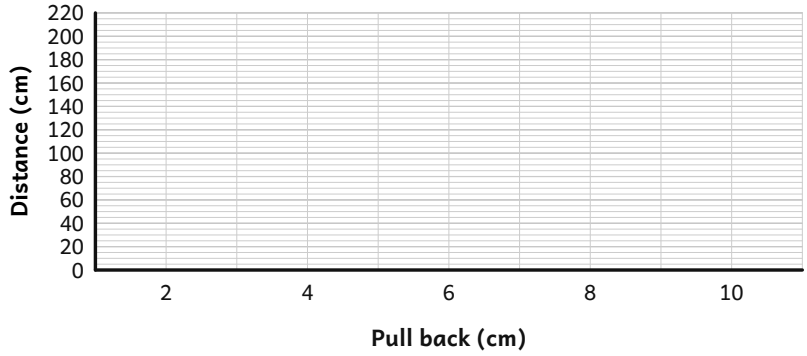
Class 4A used a rubber band stretched between two chair legs to launch a toy car. They pulled the band back by different amounts and measured how far the car travelled.

1 **Complete the plan.** [3]

Change	
Measure	
Keep the same	

2 **Draw a line graph.** [3]

Pull back (cm)	2	4	6	8	10
Distance travelled (cm)	40	85	130	170	210



3 **Describe the pattern and explain it using energy.** [3]

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.....

.....

4 **Predict the distance for a pull of 12 cm.** [1]

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REAL-WORLD CASE STUDY · C8 · ENERGY

Cooking with cow dung: biogas

★★★ Grade 4

Name: Class: Date:

Learning goals

- Track energy through a real system.
- Use numbers.
- Weigh up benefits.

Science | Farming | Health | Environment | Economics

In many villages in Bangladesh, families cook on clay stoves that burn wood, leaves or dried cow dung. The smoke fills the kitchen and harms people's lungs, especially women and children.

A **biogas plant** offers another way. Cow dung and water are fed into a closed underground tank. Tiny **microbes** break down the dung and make a gas called **methane**. The gas flows through a pipe to a stove in the kitchen, where it burns with a clean blue flame. The leftover **slurry** is an excellent fertiliser for the fields. A family with a few cows can make enough gas to cook their meals.

Based on information about household biogas in Bangladesh (e.g. IDCOL's biogas programme). Numbers below are examples.

1 **Complete the energy chain for biogas cooking.** [3]

- ☐ gas burns: heat for cooking
- ☐ light from the Sun
- ☐ microbes make biogas
- ☐ grass grows (chemical energy)
- ☐ cows eat grass; dung holds chemical energy

2 **Why is biogas called a renewable energy source?** [2]

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.....

3 **Maths. A family uses 15 kg of firewood a week. A biogas plant replaces all of it. How much wood is saved in a year (52 weeks)?** [2]

.....

4 **Give one benefit of biogas for each.** [3]

For...	Benefit
health	
the environment	
farming	

Show what you know

★★★ Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

I	S	O	R	E	H	O	Y	W	P
S	W	T	H	O	K	H	E	A	T
B	J	J	O	O	I	F	Y	C	E
I	S	K	N	R	N	I	L	H	P
O	O	E	M	U	E	O	I	A	U
G	U	N	F	S	T	D	G	I	A
A	N	E	Y	O	I	S	H	N	V
S	D	R	J	L	C	J	T	J	D
H	B	G	L	A	N	D	W	C	I
E	G	Y	S	R	P	T	K	L	J

biogas chain energy heat kinetic light solar sound stored

2 What energy change happens in a solar panel? [1]

- ◯ heat → sound
- ◯ light → electrical
- ◯ electrical → light
- ◯ kinetic → chemical

3 Which energy source will run out? [1]

- ◯ wind
- ◯ sunlight
- ◯ natural gas
- ◯ biogas

4 Draw an energy chain for a mobile phone playing music, starting from the charged battery. [3]

Phone energy chain

5 **Challenge. "Energy is never used up, it only changes form." Explain this using a bicycle that slows down and stops when you stop pedalling.** [2]

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How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name forms of energy	○	○	○
draw energy chains	○	○	○
explain energy transfer between objects	○	○	○
track energy from the Sun through a system	○	○	○
investigate stored energy in a rubber band	○	○	○
explain how biogas works and why it is renewable	○	○	○